

Project 1: Intake Piping

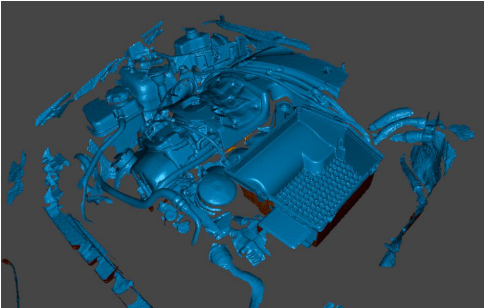
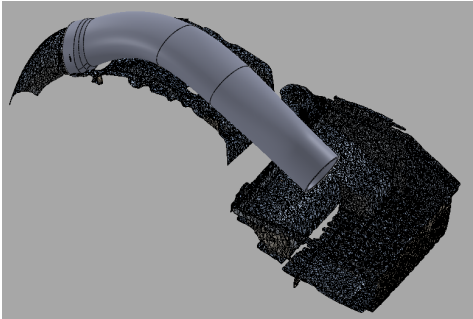
Reverse-engineering workflow using 3D scanning, mesh preparation, CAD design, FDM prototyping and vehicle test fitting.

3D SCANNING | MESH PROCESSING | CAD MODELLING | FDM PRINTING | FITMENT VALIDATION

Project output

Factory intake was incompatible with generic cone filters; replacements were back-ordered from Europe. Custom piping was reverse-engineered from a 3D scan to accept a 76mm cone filter — the largest filter OD that clears the bonnet and factory airbox while retaining factory air routing within the scanned envelope.

SCANNING, MESH PROCESSING AND CAD DESIGN

PART 1 Scanning the area  Scanned the engine bay to capture packaging space, clearances and surrounding components. Physical measurements were taken alongside the scan to verify geometrical accuracy and ensure the captured data reflected real dimensions before progressing to mesh processing.	PART 2 Processing the mesh  Multiple scans were completed and the cleanest result — free of artefacts, with clear geometry and undisrupted surfaces — was selected. The mesh was then processed by progressively removing areas irrelevant to the required geometry until only the needed sections remained. Resolution was refined to a workable file size before being imported into SolidWorks as the design reference.	PART 3 Designing the inlet pipe  Modelled the inlet pipe directly around the 3D scan geometry to ensure clear routing without potential interference with surrounding components. Clearances were built in from the start rather than assumed, which was confirmed during later test fitting.
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DESIGN FOCUS

Scan quality selection Multiple captures were taken and the cleanest scan chosen to ensure the downstream mesh and CAD work was based on accurate, reliable geometry from the start.	Scan-referenced geometry Modelling directly around the scan embedded real packaging constraints into the design, eliminating assumptions about clearances and reducing the risk of interference during fitment.	Connector-first workflow Printing and validating the most constrained interface first reduced rework risk and created a manageable, staged workflow for the remaining pipe sections.
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Prototyping, fitment validation and outcome.

3D SCANNING | MESH PROCESSING | CAD MODELLING | FDM PRINTING | FITMENT VALIDATION

Project output

Custom piping accepting a 76mm cone filter, mating to the factory oval outlet (105 × 63 mm OD) with a minimum 15mm clearance to the engine cover. FDM sections are acetone solvent-welded and RTV-sealed at the joints. The packaging-driven 10% convergence uses a smooth transition to limit loss; an airflow test against the OEM pipe is the next step. Final part in ABS for underbonnet heat and chemical resistance.

PROTOTYPING, FITMENT AND OUTCOME

PART 4 Printing the connector and test fitting



Critical interface printed and fit-checked first: the connector receives the factory male outlet — an oval section, 105 × 63 mm OD — reverse-engineered from the scan with a [X] mm slip-fit allowance for acetone bonding. Resolving the most constrained geometry early de-risked the remaining sections.

PART 5 Printing the remaining piping



With the interface confirmed, the remaining sections were printed, dry-assembled, then joined by acetone solvent-welding into a single body, with the slip joints sealed using RTV silicone. Staged printing kept each section manageable and editable without reprinting the full assembly.

PART 6 Test fitting on the vehicle



Installed on the vehicle with a minimum 15mm clearance to the engine cover — verified with a 15mm flexible shaft at the tightest point — and no interference, validating the scan-referenced design. The packaging-driven 10% convergence uses a smooth transition to limit pressure loss; a back-to-back pressure test against the OEM pipe is the next step.

OUTCOME FOCUS

Fitment validated by scan

A minimum 15mm clearance to the engine cover and no interference at first fit — designing directly around the scan produced an installation-ready part without iteration.

Airflow — loss control

The 10% area convergence is a packaging constraint, not a performance feature; a smooth converging transition limits the resulting loss. Its effect against the OEM pipe will be quantified by a back-to-back turbo-inlet pressure test (pending).

Next steps and material selection

Final part in ABS: a glass-transition temperature near 100°C suits the underbonnet environment (versus nylon’s ~70°C), and acetone solvent-welding gives strong, sealed joints between the FDM sections. ASA offers better UV and heat-soak resistance but bonds less readily with acetone — if chosen, joint strength must be re-validated.